

TBI Technical Team — Selection Task

What this is for

This isn't about producing polished code — it's about whether your fix actually works under the edge cases below, and whether you can explain your own reasoning. In the interview, we'll ask you to walk through your submission live.

Part 1 — Fix the bug

Below is a countdown timer meant for on-stage use. It has more than one bug. Find and fix them.

```
<div id="timer">10:00</div>
<button id="startBtn">Start</button>
<script>
  let timeLeft = 600; // 10 minutes in seconds
  function startTimer() {
    setInterval(() => {
      timeLeft--;
      let minutes = Math.floor(timeLeft / 60);
      let seconds = timeLeft % 60;
      document.getElementById('timer').innerText = minutes + ":" + seconds;
    }, 1000);
  }
  document.getElementById('startBtn').addEventListener('click', startTimer);
</script>
```

Hint: click Start more than once and watch what happens. Also watch what the display looks like once seconds drops below 10, and what happens when it reaches zero.

Part 2 — Add a feature

The organizing team wants to reuse this timer for Demo Day, with two requirements:

- An admin should be able to pause and resume the countdown mid-run (e.g. via a button).
- The countdown must stay correct even if the laptop running it sleeps, the browser tab loses focus for a while, or the system clock is slightly off when it wakes back up — it should not silently drift or lose time.

You may restructure the code however you want to meet these requirements. There's no single correct approach — we're evaluating your reasoning as much as the result.

Part 3 — Submit

Your working code — in a GitHub repo.

Structure:

timer-fix/

```
|— README.md    # Candidate details, solution approach, key decisions, and reasoning
|— original.html # the buggy snippet, untouched, for easy side-by-side diffing
|— part1.html   # bug-fixed version (Part 1 only)
└— part2.html   # part1 + pause/resume + drift-proofing (final version)
```

You can use AI but don't submit something you can't explain. We will ask you to justify a specific line or decision live in the interview.

We look forward to discussing your approach with you during the interview.

The Documentation Team has been merged with the Tech Team, so candidates with strong data-handling skills may receive some consideration in cases where their coding skills are comparatively weaker.

**All the best,
Shashank Shekhar**